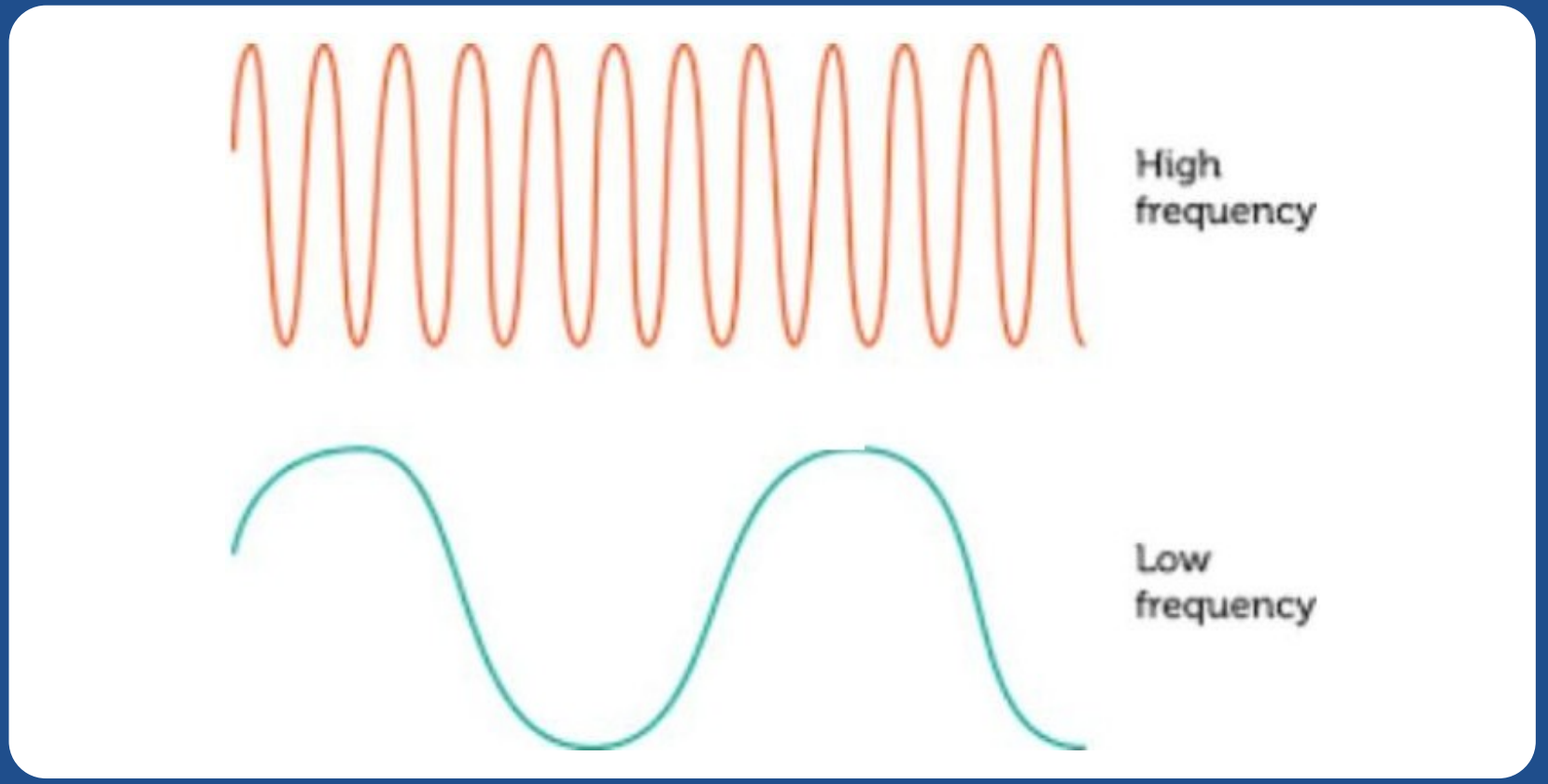
4.03 Frequency and Speed of a Wave

How are frequency, wavelength, and wave speed connected?



Today's Goals

I can define frequency and measure it in hertz (Hz).

I can explain: high frequency = short wavelength.

I can explain why frequency does NOT change wave speed.

I can explain how the medium changes wave speed.

Rate each goal:

1 Not yet

2 Getting there

3 I can teach it

Ohio Science Standards

7.PS.4 Energy can be transferred through a variety of ways.

- A wave travels at a constant speed through a material as long as the material is uniform.
- The speed of a wave depends on the material (sound travels faster through most solids than gases).
- Waves are described by speed, wavelength, amplitude, and frequency.

Inquiry skills (Grades 5–8): "Use appropriate mathematics, tools and techniques to gather data" and "Think critically and logically to connect evidence and explanations" (today's CER lab).

Source: Ohio's Learning Standards for Science (2018), Grade 7 Physical Science; content statement paraphrased.

In this lesson

- Integer warm-up
- Vocab check-in
- Frequency and hertz
- Frequency and wavelength
- Frequency does not change speed
- The medium changes speed
- CER lab + Quest Map

CASE #4 • THURSDAY

Is the dolphin at its pod's depth?

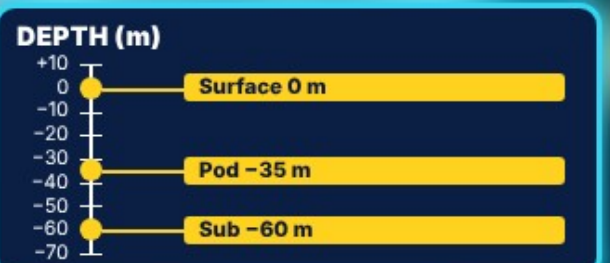
WAVE WORD: *Depth*
The distance below the surface.

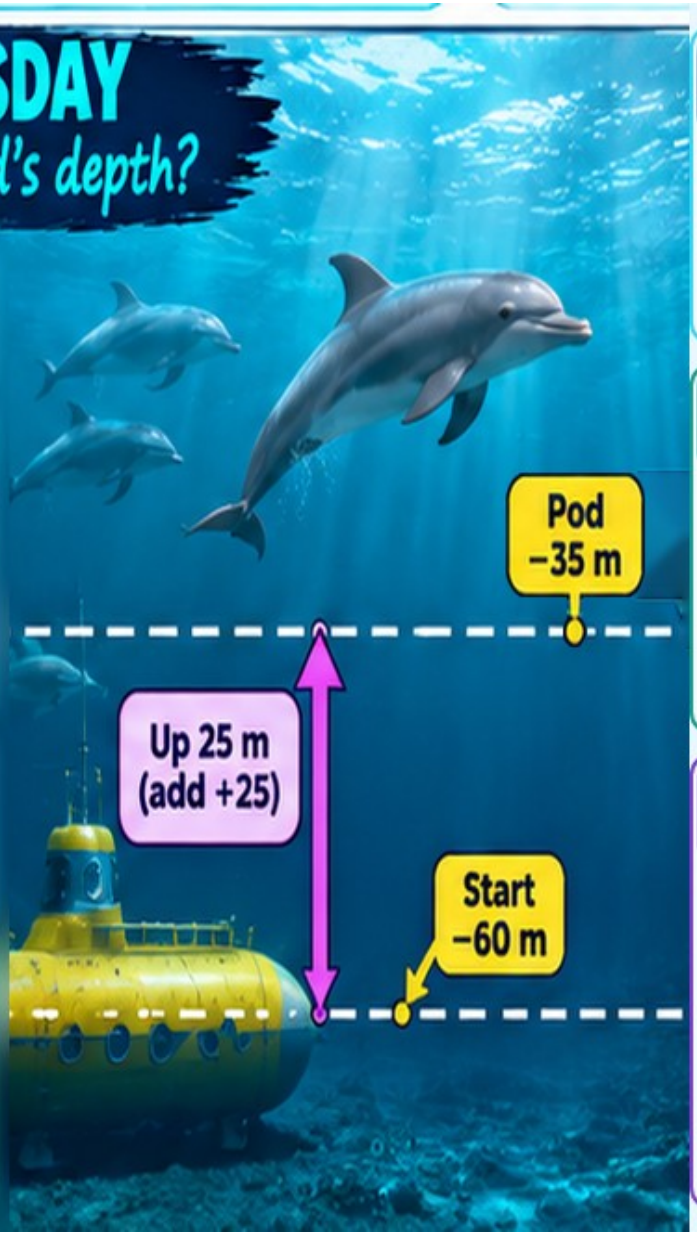
MYSTERY

A dolphin swims by a submarine 60 meters below the surface (−60 m). It swims up 25 m. Its pod swims 35 meters below the surface (−35 m). Is the dolphin at its pod's depth?

CLUES

- Start (by the sub): −60 m
- Move up: +25 m
- Pod's depth: −35 m





CLAIM

Is the dolphin at its pod's depth?

☐ YES ☐ NO

EVIDENCE

Solve:

−60 + 25 = _____

The dolphin is now at _____ m.

REASONING

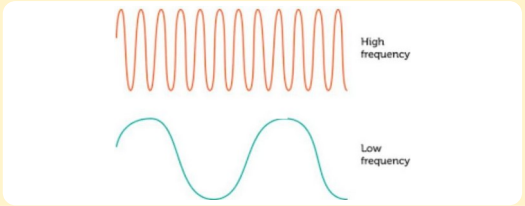
Finish the sentence:

The signs are different, so I subtract 60 − 25 = _____ and keep the sign of _____.

The dolphin is at _____ m, the _____ as its pod.

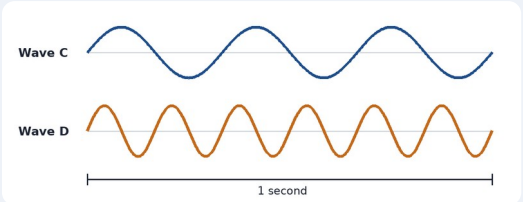
Vocab Check-In

From your Unit 4 vocabulary pretest. Today we focus on the starred words.



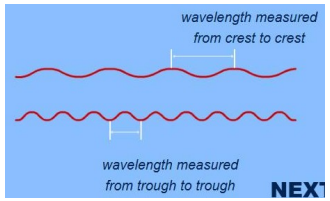
frequency ★

the number of waves that go past a point in a given time



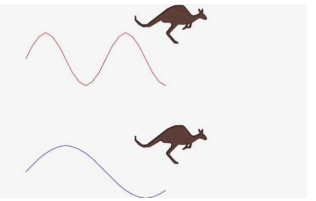
hertz (Hz)

the unit for frequency: 1 Hz = 1 wave per second



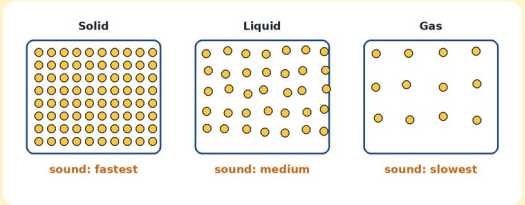
wavelength ★

the distance from one crest to the next crest



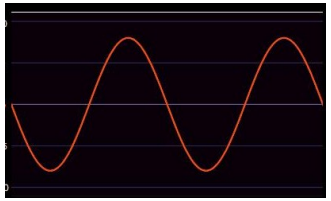
wave speed

how far a wave travels in a given time: speed = distance ÷ time



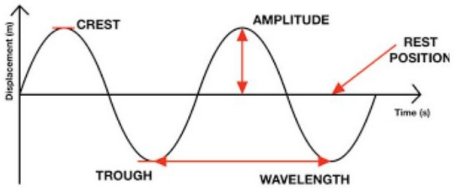
medium ★

the material a wave travels through (rope, water, air, steel)



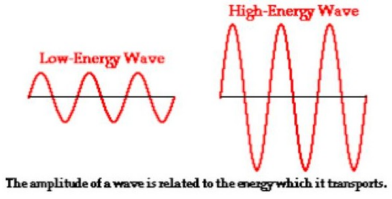
wave

a disturbance that carries energy from one place to another



amplitude

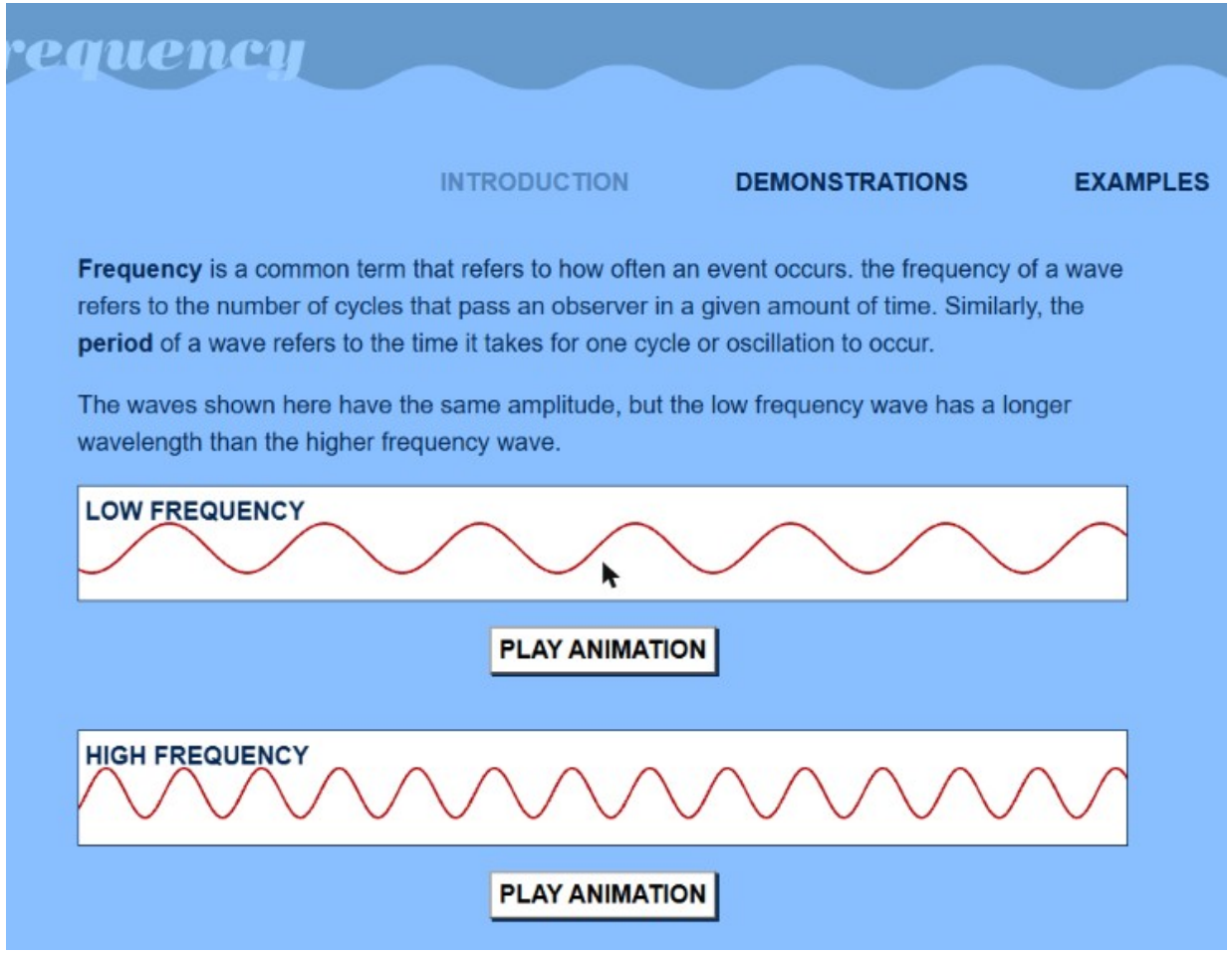
how far a wave moves from its rest position



energy

the ability to do work or cause change

What Is Frequency?



Frequency = the number of waves that go past a point in a given time.

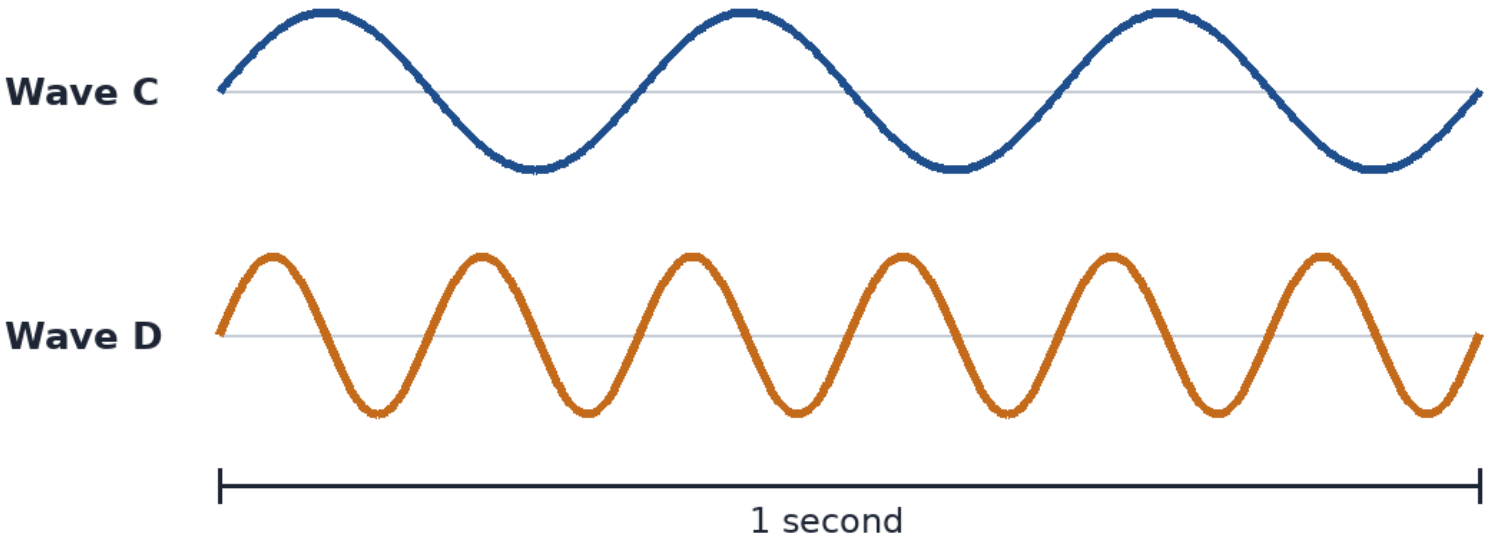
It is measured in **hertz (Hz)**. 1 Hz = 1 wave per second.

Raise the frequency: more waves pass each second, and the waves get **closer together**.

▶️ PBS: Exploring Frequency

Classkick: 4.03 Part 1 · Q2

Count the Waves



Wave C: ___ Hz

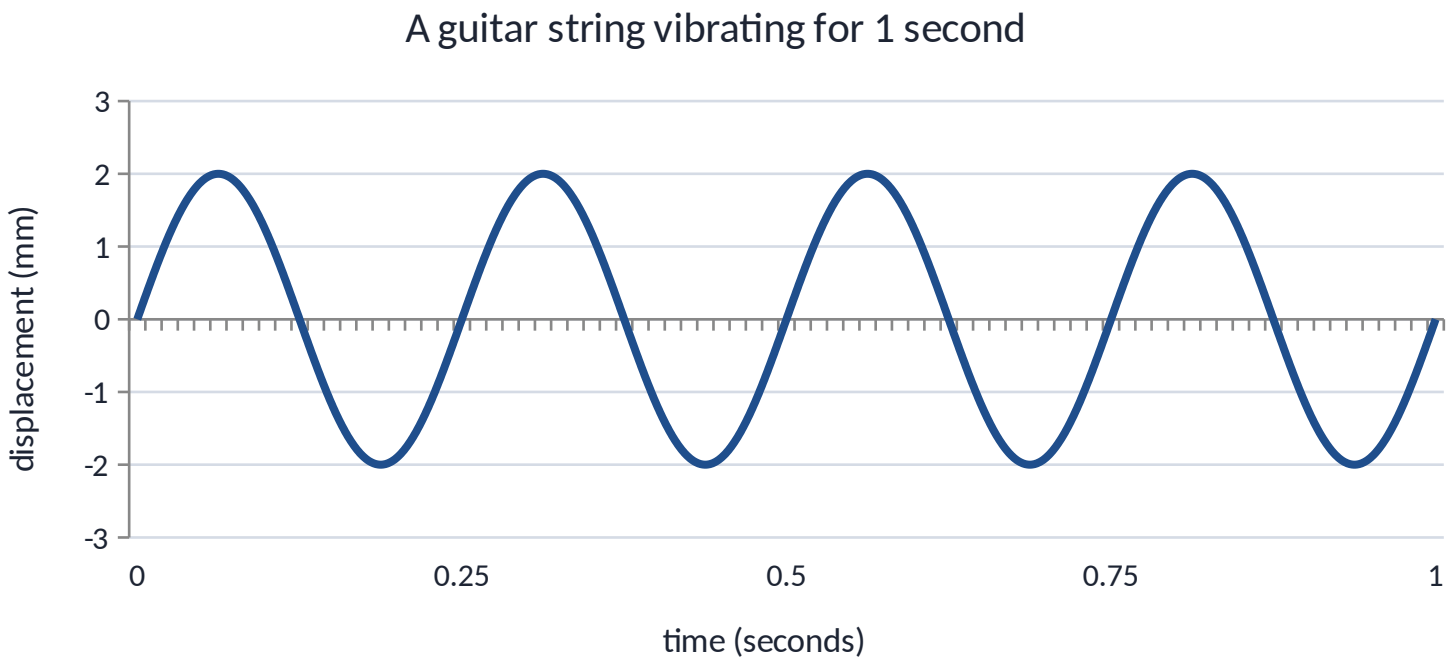
Wave D: ___ Hz

Higher frequency? C / D

Shorter wavelength? C / D

One cycle = one full up and one full down. We do this one together; you do Q2 on your Classkick.

Graph Poll



Read the graph.

What is the frequency of this wave?

Hint: count full cycles in 1 second.

Single choice poll

A. 2 Hz

Single choice poll

B. 4 Hz

Single choice poll

C. 8 Hz

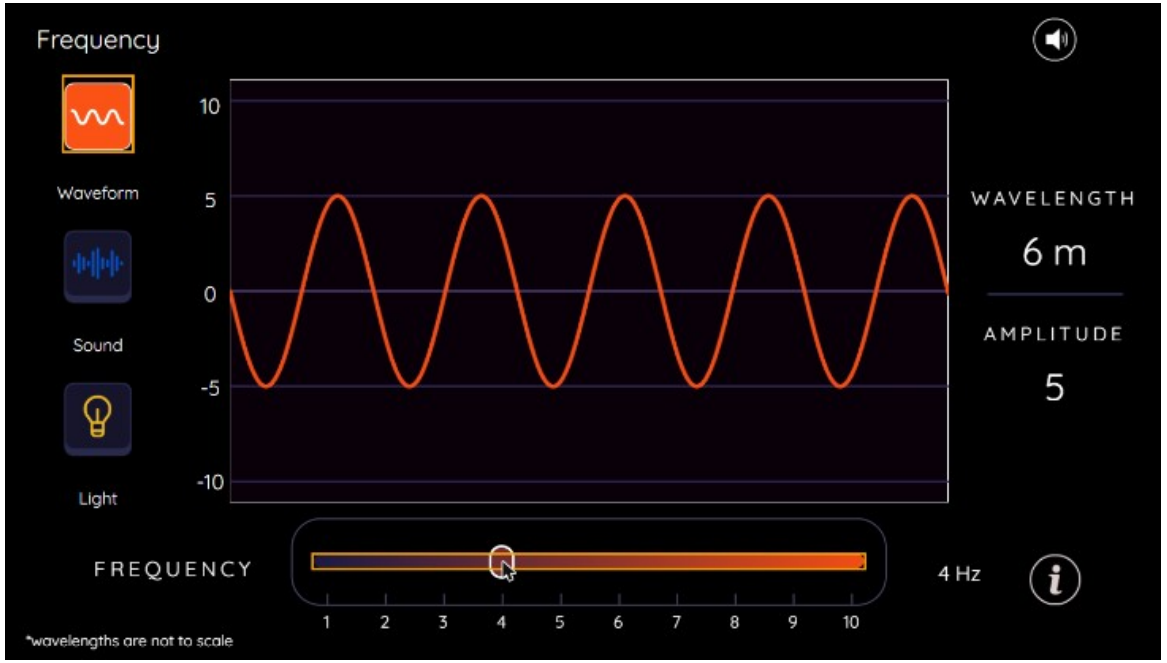
Single choice poll

D. 1 Hz

Frequency and Wavelength Go Together



Low frequency (1 Hz): long wavelength



High frequency (4 Hz): short wavelength

High frequency = short wavelength. Low frequency = long wavelength.
The amplitude did not change.

▶ **PBS: Frequency and Wavelength**

Video: Wavelength, Frequency, and Amplitude



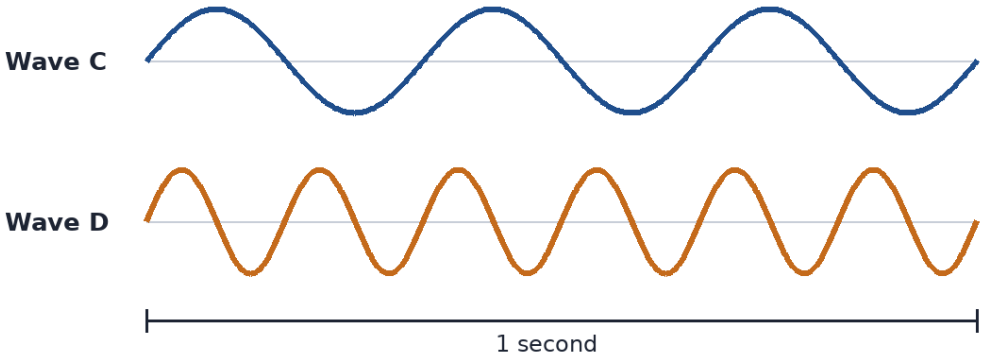
Knowledge Platform (plays right here in Slide Show)

Listen for:

- What does a high-pitched sound do to the frequency?
- What happens to the wavelength when frequency goes up?
- What does amplitude change: loudness or pitch?

▶️ Backup: YouTube

Engageli Poll



Wave D has a higher frequency than Wave C. Compared to Wave C, Wave D has a...

Single choice poll

A. longer wavelength

Single choice poll

B. shorter wavelength

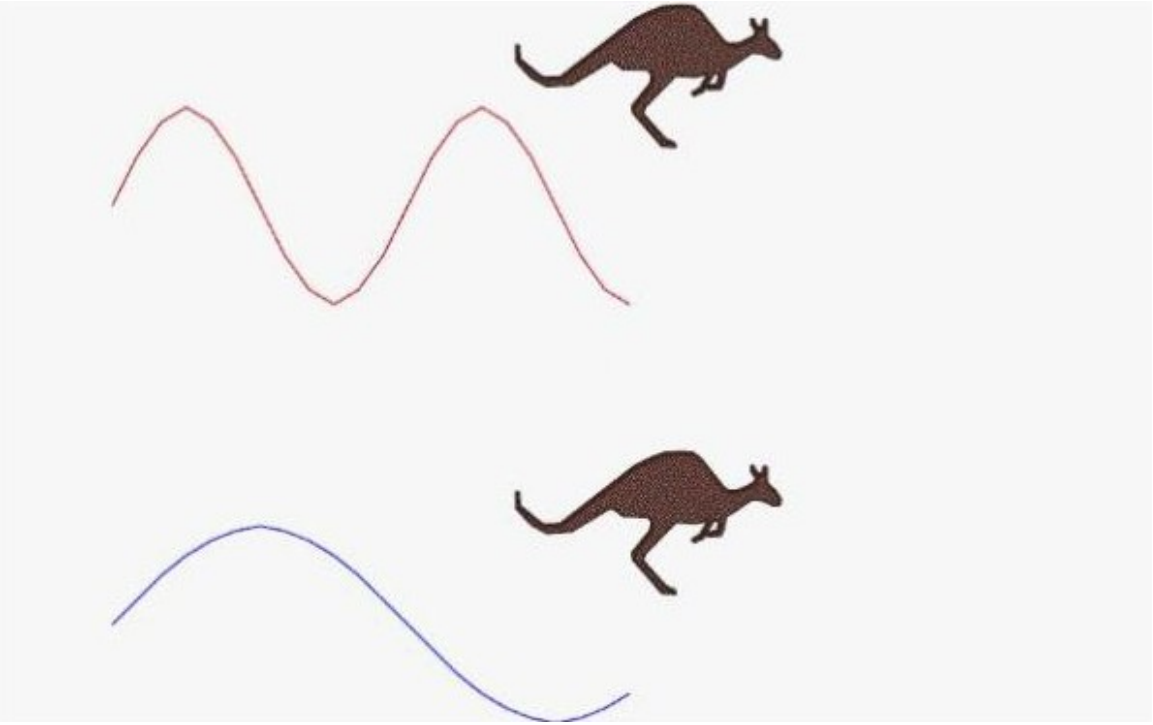
Single choice poll

C. faster speed

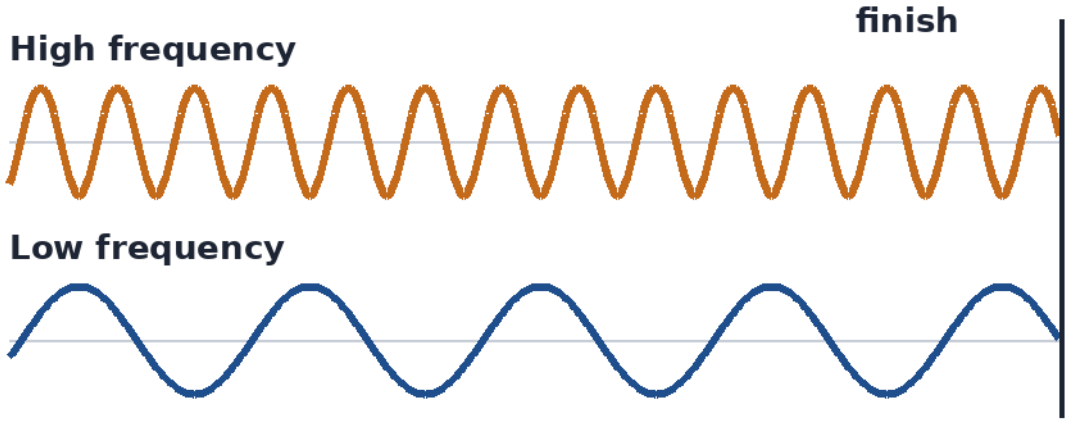
Single choice poll

D. bigger amplitude

Frequency Doesn't Change Speed



Both kangaroos reach the end at the same time.



Watch: both waves finish together.

Remember: in the same medium, high and low frequency waves travel at the SAME speed. Wave speed depends on the medium, not the frequency.

Who Is Right?

A high whistle and a low drum sound at the same moment across the gym. Who explains it best?

Ava says:

"The whistle reaches you first because high-frequency waves are faster."

Ben says:

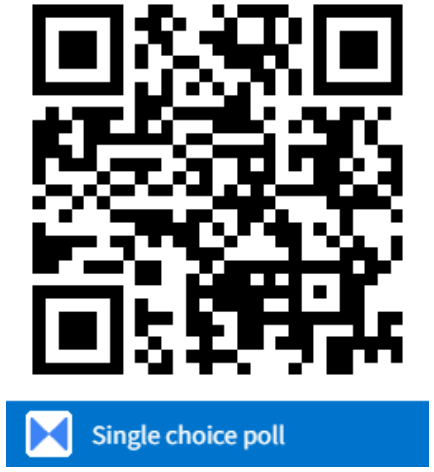
"The drum reaches you first because its waves are bigger and stronger."

Cam says:

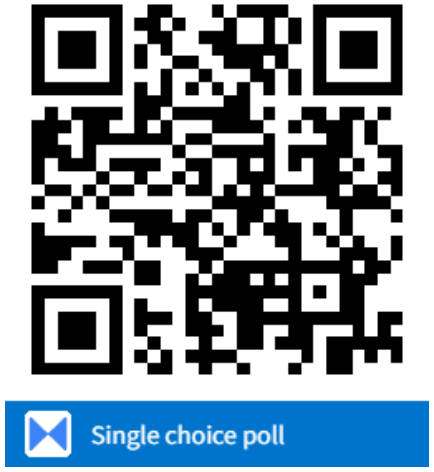
"They arrive at the same time because both travel through the same air."

Dee says:

"The drum wins because long wavelengths travel farther each second."



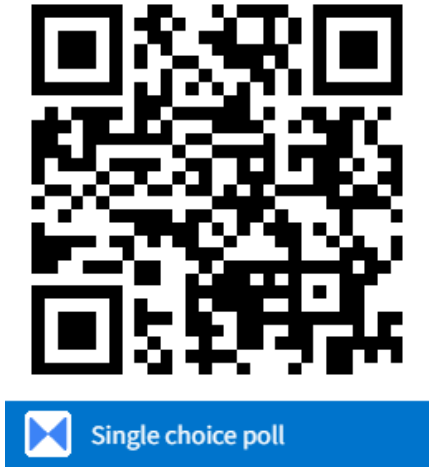
A. Ava



B. Ben

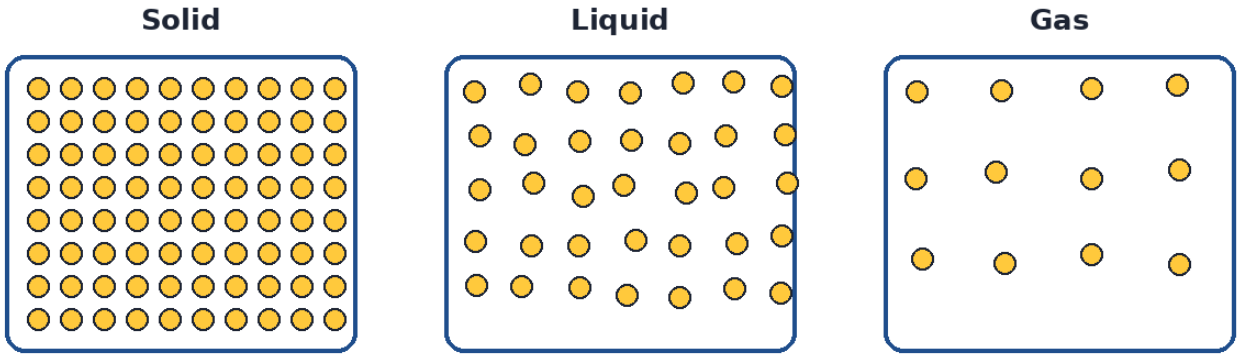


C. Cam



D. Dee

The Medium Changes the Speed



Particles close together pass the energy along faster. Sound is usually fastest in solids and slowest in gases.

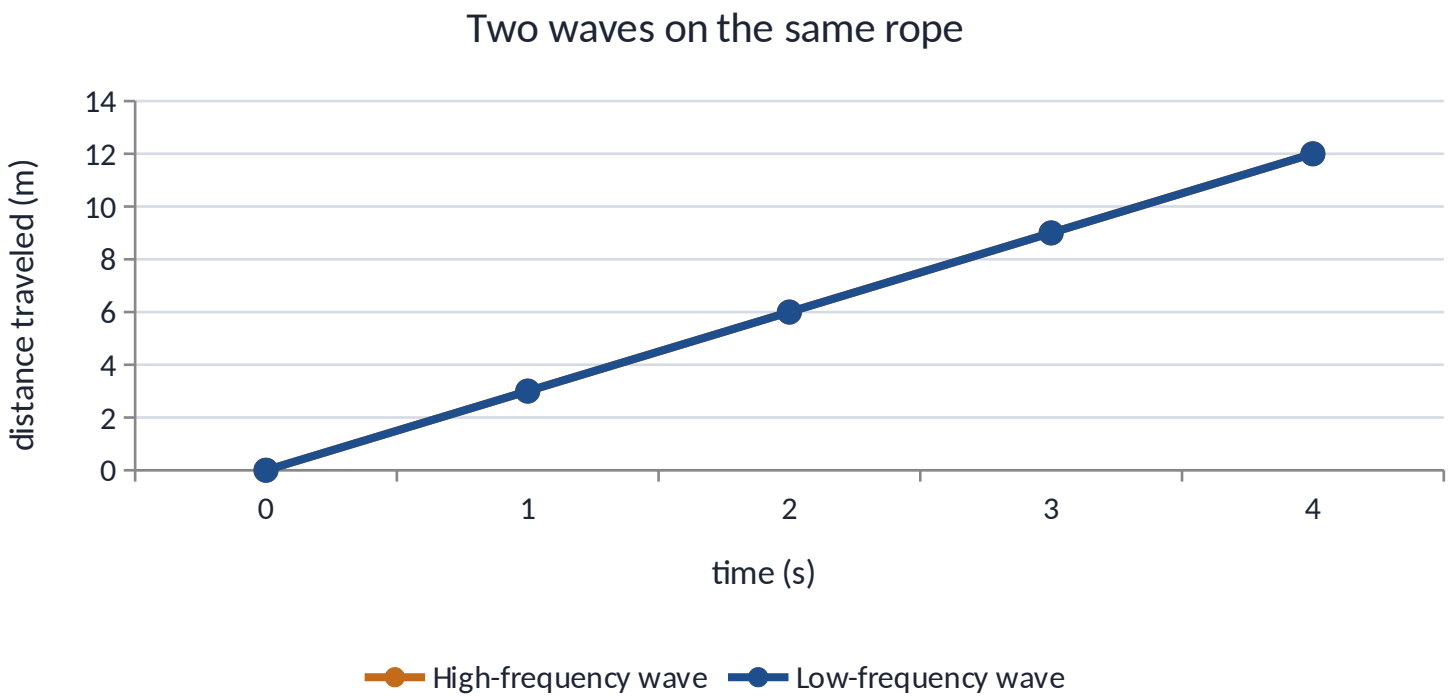


Same medium: same speed



Two mediums: two speeds

Graph Poll



Read the graph.

What does the graph show about the two waves?

Tip: the lines are on top of each other.

Single choice poll

A. Same speed: 3 m/s

Single choice poll

B. High frequency is faster

Single choice poll

C. Low frequency is faster

Single choice poll

D. Neither wave moves

CER Lab: Frequency and Wavelength

Our question: "Does increasing the frequency of a wave on a string affect its wavelength? If so, how?"

1. Type your name, then click CONTINUE.
2. Set up PhET: damping none, tension high, No End, Oscillate, ruler.
3. Measure crest to crest at 1.0, 1.5, 2.0, 2.5, and 3.0 Hz.
4. Write your Claim, Evidence, and Reasoning from YOUR data.
5. Screenshot your CER and paste it on your Classkick.

▶ **Open the CER lab**

Click the blue ? on the website for sentence starters. Not done? Finish it on Friday.

Get Ready for the 4.03 Quest

Lesson 4.01

Crests are the high points. After the wave passes, the rope returns to its resting position.

Lesson 4.01

Wavelength = crest to crest. Troughs below zero have negative numbers.

Lesson 4.02

Bigger amplitude = more energy.

Lesson 4.03

Higher frequency = shorter wavelength. Same medium = same speed.

4.03 Quest (10 pts): due Thu, Nov 19 (extended time: Mon, Nov 30). Use your Quest Map!

▶️ **Waves Review games (Wordwall)**

Key Ideas

1. Frequency = waves per second, measured in hertz (Hz).
2. High frequency = short wavelength. Low frequency = long wavelength.
3. In the same medium, all frequencies travel at the same speed.
4. The medium changes the speed: sound is usually fastest in solids.

Re-rate your goals: 1 = Not yet, 2 = Getting there, 3 = I can teach it